

1. Let V be a real inner product space. A set $S = \{\mathbf{u}_1, \dots, \mathbf{u}_k\} \subseteq V$ is called *orthogonal* if it doesn't contain the zero vector, and if

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0 \text{ for all } i \neq j.$$

(a) Let $S = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ be an orthogonal set and suppose that

$$\mathbf{v} = a_1 \mathbf{u}_1 + a_2 \mathbf{u}_2 + \dots + a_k \mathbf{u}_k.$$

In this case, show that $a_i = \langle \mathbf{v}, \mathbf{u}_i \rangle / \langle \mathbf{u}_i, \mathbf{u}_i \rangle$ for all i . [Remark: If we also have $\langle \mathbf{u}_i, \mathbf{u}_i \rangle = 1$ for all i then the set S is called *orthonormal*, in which case we have $a_i = \langle \mathbf{v}, \mathbf{u}_i \rangle$.]

(b) Use part (a) to prove that every orthogonal set is an independent set.

(a): Fix some $1 \leq i \leq k$ and the inner product of each side with $\mathbf{u}_i$ to get

$$\begin{aligned} \langle \mathbf{v}, \mathbf{u}_i \rangle &= \langle a_1 \mathbf{u}_1 + a_2 \mathbf{u}_2 + \dots + a_k \mathbf{u}_k, \mathbf{u}_i \rangle \\ &= a_1 \langle \mathbf{u}_1, \mathbf{u}_i \rangle + a_2 \langle \mathbf{u}_2, \mathbf{u}_i \rangle + \dots + a_k \langle \mathbf{u}_k, \mathbf{u}_i \rangle \\ &= a_i \langle \mathbf{u}_i, \mathbf{u}_i \rangle. \end{aligned}$$

Then divide both sides by $\langle \mathbf{u}_i, \mathbf{u}_i \rangle$, which is possible because $\mathbf{u}_i \neq \mathbf{0}$.

(b): Suppose that $a_1 \mathbf{u}_1 + \dots + a_k \mathbf{u}_k = \mathbf{0}$. Then part (a) tells us that

$$a_i = \frac{\langle \mathbf{0}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} = \frac{0}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} = 0 \quad \text{for all } i.$$

2. Let W be a vector space and let $U, V \subseteq W$ be subspaces.

(a) Prove that $U \cap V$ is a subspace, but $U \cup V$ need not be a subspace.

(b) Define the sum of subspaces by

$$U + V = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in U, \mathbf{v} \in V\}.$$

Prove that $U + V$ is a subspace of W .

(c) Suppose that $U \cap V = \{\mathbf{0}\}$. In this case, show that for all vectors $\mathbf{w} \in U + V$ there exists a **unique** expression $\mathbf{w} = \mathbf{u} + \mathbf{v}$ with $\mathbf{u} \in U$ and $\mathbf{v} \in V$. [Hint: Suppose that $\mathbf{u} + \mathbf{v} = \mathbf{u}' + \mathbf{v}'$ with $\mathbf{u}, \mathbf{u}' \in U$ and $\mathbf{v}, \mathbf{v}' \in V$. Then ...]

(d) If $W = U + V$ and $U \cap V = \{\mathbf{0}\}$ then we say that W is the *direct sum* of U and V , and we write

$$W = U \oplus V.$$

In this case, show that $\dim W = \dim U + \dim V$. [Hint: Let $\mathbf{u}_1, \dots, \mathbf{u}_m$ be a basis of U and let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be a basis of V . Prove that $\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_1, \dots, \mathbf{v}_n$ is a basis of W .]

(a): To show that $U \cap V$ is a subspace, consider any $\mathbf{x}, \mathbf{y} \in U \cap V$ and $a, b \in \mathbb{R}$. Since $U \cap V \subseteq U$ we have $\mathbf{x}, \mathbf{y} \in U$, hence $a\mathbf{x} + b\mathbf{y} \in U$ because U is a subspace. Similarly, since $U \cap V \subseteq V$ and since V is a subspace we have $a\mathbf{x} + b\mathbf{y} \in V$. Hence $a\mathbf{x} + b\mathbf{y} \in U \cap V$, as desired. To show that $U \cup V$ is **not** necessarily a subspace, pick any two linearly independent vectors $\mathbf{x}, \mathbf{y} \in W$, and consider the lines $U = \text{Span}\{\mathbf{x}\} = \{a\mathbf{x} : a \in \mathbb{R}\}$ and $V = \text{Span}\{\mathbf{y}\} = \{b\mathbf{y} : b \in \mathbb{R}\}$, generated by $\mathbf{x}$ and $\mathbf{y}$. Then the vector $\mathbf{x} + \mathbf{y}$ is not in U because this would imply $(1 - a)\mathbf{x} + \mathbf{y} = \mathbf{0}$ and it is not in V since this would imply $\mathbf{x} + (1 - b)\mathbf{y} = \mathbf{0}$, contradicting the independence of $\mathbf{x}$ and $\mathbf{y}$. Hence $\mathbf{x} + \mathbf{y}$ is not in $U \cup V$.

(b): Consider any vectors $\mathbf{x}, \mathbf{y} \in U + V$ and scalars $a, b \in \mathbb{R}$. By definition we can write $\mathbf{x} = \mathbf{u} + \mathbf{v}$ and $\mathbf{y} = \mathbf{u}' + \mathbf{v}'$ for some $\mathbf{u}, \mathbf{u}' \in U$ and $\mathbf{v}, \mathbf{v}' \in V$. Then then we have

$$a\mathbf{x} + b\mathbf{y} = a(\mathbf{u} + \mathbf{u}') + b(\mathbf{v} + \mathbf{v}') = (a\mathbf{u} + b\mathbf{u}') + (a\mathbf{v} + b\mathbf{v}'),$$

which is in $U + V$ because $a\mathbf{u} + b\mathbf{u}' \in U$ and $a\mathbf{v} + b\mathbf{v}' \in V$.

(c): Suppose that $U \cap V = \{\mathbf{0}\}$ and suppose we have $\mathbf{u} + \mathbf{v} = \mathbf{u}' + \mathbf{v}'$ for some $\mathbf{u}, \mathbf{u}' \in U$ and $\mathbf{v}, \mathbf{v}' \in V$. This implies that $\mathbf{u} - \mathbf{u}' = \mathbf{v}' - \mathbf{v}$. But $\mathbf{u} - \mathbf{u}' \in U$ and $\mathbf{v}' - \mathbf{v} \in V$, which implies that $\mathbf{u} - \mathbf{u}'$ and $\mathbf{v}' - \mathbf{v}$ are both in $U \cap V$. Since $U \cap V = \{\mathbf{0}\}$ this implies that $\mathbf{u} - \mathbf{u}' = \mathbf{0}$, hence $\mathbf{u} = \mathbf{u}'$, and $\mathbf{v}' - \mathbf{v} = \mathbf{0}$, hence $\mathbf{v} = \mathbf{v}'$.

(d): Suppose that $W = U + V$ and $U \cap V = \{\mathbf{0}\}$. Let $\mathbf{u}_1, \dots, \mathbf{u}_m$ be a basis of U and let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be a basis of V . I claim that $B = \{\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis of W consisting of $m + n$ distinct vectors, and hence

$$\dim W = m + n = \dim U + \dim V.$$

To see that the vectors are distinct, suppose that $\mathbf{u}_i = \mathbf{v}_j$ for some i, j . But then $\mathbf{u}_i, \mathbf{v}_j$ are both in $U \cap V = \{\mathbf{0}\}$, which implies $\mathbf{u}_i = \mathbf{0}$ and $\mathbf{v}_j = \mathbf{0}$. But no basis can contain the zero vector. To show that B is a spanning set, note that for any $\mathbf{w} \in W$ there exist $\mathbf{u} \in U$ and $\mathbf{v} \in V$ such that $\mathbf{w} = \mathbf{u} + \mathbf{v}$ because we have $W = U + V$ by assumption. Since $\mathbf{u}_1, \dots, \mathbf{u}_m$ is a basis for U and $\mathbf{v}_1, \dots, \mathbf{v}_n$ is a basis for V , there exist scalars a_i and b_j such that

$$\mathbf{w} = \mathbf{u} + \mathbf{v} = (a_1\mathbf{u}_1 + \dots + a_m\mathbf{u}_m) + (b_1\mathbf{v}_1 + \dots + b_n\mathbf{v}_n).$$

To show independence, suppose we have

$$a_1\mathbf{u}_1 + \dots + a_m\mathbf{u}_m + b_1\mathbf{v}_1 + \dots + b_n\mathbf{v}_n = \mathbf{0},$$

for some scalars a_i and b_j . This implies that

$$a_1\mathbf{u}_1 + \dots + a_m\mathbf{u}_m = -b_1\mathbf{v}_1 - \dots - b_n\mathbf{v}_n.$$

But the left hand side is a vector in U and the right hand side is a vector in V . Hence each side is in $U \cap V = \{\mathbf{0}\}$, which implies that each side equals $\mathbf{0}$. Then the independence of the $\mathbf{u}_i$ implies that $a_i = 0$ for all i , and the independence of the $\mathbf{v}_j$ implies that $b_j = 0$ for all j .

3. Let V be a real inner product space and let $U \subseteq V$ be a subspace. We define the *orthogonal complement* of U as follows:

$$U^\perp := \{\mathbf{v} \in V : \langle \mathbf{v}, \mathbf{u} \rangle = 0 \text{ for all } \mathbf{u} \in U\}.$$

- (a) Prove that $U^\perp$ is a subspace of V .
- (b) Prove that $U \cap U^\perp = \{\mathbf{0}\}$.
- (c) If V is finite dimensional, prove that $V = U + U^\perp$. Combining this with (b) gives

$$V = U \oplus U^\perp,$$

and hence $\dim V = \dim U + \dim U^\perp$. [Hint: Suppose that $\dim U = k$ and $\dim V = n$, so that $k \leq n$. From the Gram-Schmidt procedure (see below), it is possible to find an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_n$ for V such that $\mathbf{u}_1, \dots, \mathbf{u}_k$ is a basis for U . Show that $\mathbf{u}_{k+1}, \dots, \mathbf{u}_n$ are in $U^\perp$ and use this to show that every $\mathbf{v} \in V$ can be expressed as $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$ and $\mathbf{w} \in U^\perp$.]

Consider any $\mathbf{x}, \mathbf{y} \in U^\perp$ and $a, b \in \mathbb{R}$. Then for all $\mathbf{u} \in U$ we have

$$\langle a\mathbf{x} + b\mathbf{y}, \mathbf{u} \rangle = a\langle \mathbf{x}, \mathbf{u} \rangle + b\langle \mathbf{y}, \mathbf{u} \rangle = a0 + b0 = 0,$$

which implies that $a\mathbf{x} + b\mathbf{y} \in U^\perp$.

(b): Suppose that $\mathbf{x} \in U \cap U^\perp$. Since $\mathbf{x} \in U^\perp$ we have $\langle \mathbf{x}, \mathbf{u} \rangle = 0$ for all $\mathbf{u} \in U$. But since $\mathbf{x} \in U$ we can take $\mathbf{u} = \mathbf{x}$ to obtain $\langle \mathbf{x}, \mathbf{x} \rangle = 0$. This implies $\mathbf{x} = \mathbf{0}$ by the axioms of inner products.

(c): From the hint, we may assume that there exists an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_n$ for V such that $\mathbf{u}_1, \dots, \mathbf{u}_k$ is an orthonormal basis for U , for some $0 \leq k \leq n$.¹ For any $1 \leq i \leq k$ and $k+1 \leq j \leq n$ we have $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ by the assumed orthogonality of the basis. This implies that each $\mathbf{u}_j$ with $k+1 \leq j \leq n$ is orthogonal to each basis vector of U , hence is orthogonal to every vector in U because

$$\left\langle \sum_{i=1}^k a_i \mathbf{u}_i, \mathbf{u}_j \right\rangle = \sum_{i=1}^k a_i \langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0.$$

The set $\mathbf{u}_1, \dots, \mathbf{u}_n$ spans V by assumption, hence for any $\mathbf{v} \in V$ we can write

$$\mathbf{v} = \sum_{i=1}^n a_i \mathbf{u}_i = \sum_{i=1}^k a_i \mathbf{u}_i + \sum_{j=k+1}^n a_j \mathbf{u}_j,$$

where $\sum_{i=1}^k a_i \mathbf{u}_i \in U$ and $\sum_{j=k+1}^n a_j \mathbf{u}_j \in U^\perp$. Hence every $\mathbf{v} \in V$ is in $U + U^\perp$.

4. Let V be a finite-dimensional real inner product space and let $U \subseteq V$ be a subspace. From Problem 3(c) we know that $V = U \oplus U^\perp$. Hence from Problem 2(c) we know that for every vector $\mathbf{x} \in V$ there exists a unique pair of vectors $\mathbf{x}_U \in U$ and $\mathbf{x}_{U^\perp} \in U^\perp$ such that

$$\mathbf{x} = \mathbf{x}_U + \mathbf{x}_{U^\perp}.$$

Define two functions $P_U, P_{U^\perp} : V \rightarrow V$, called the *orthogonal projections* onto U and $U^\perp$, by setting $P_U(\mathbf{x}) := \mathbf{x}_U$ and $P_{U^\perp}(\mathbf{x}) := \mathbf{x}_{U^\perp}$. Let $\mathbf{u}_1, \dots, \mathbf{u}_k$ be an orthogonal basis for U . In this case, show that

$$P_U(\mathbf{x}) = \frac{\langle \mathbf{x}, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1 + \frac{\langle \mathbf{x}, \mathbf{u}_2 \rangle}{\langle \mathbf{u}_2, \mathbf{u}_2 \rangle} \mathbf{u}_2 + \dots + \frac{\langle \mathbf{x}, \mathbf{u}_k \rangle}{\langle \mathbf{u}_k, \mathbf{u}_k \rangle} \mathbf{u}_k$$

and

$$P_{U^\perp}(\mathbf{x}) = \mathbf{x} - P_U(\mathbf{x}) = \mathbf{x} - \sum_{i=1}^k \frac{\langle \mathbf{x}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \mathbf{u}_i.$$

[Hint: Use Problem 1(a).]

Since $\mathbf{u}_1, \dots, \mathbf{u}_k$ is a basis of U we can write $\mathbf{x}_U = a_1 \mathbf{u}_1 + \dots + a_k \mathbf{u}_k$ for some scalars a_i , and Problem 1(a) tells us that $a_i = \langle \mathbf{x}_U, \mathbf{u}_i \rangle / \langle \mathbf{u}_i, \mathbf{u}_i \rangle$. But since $\langle \mathbf{x}_{U^\perp}, \mathbf{u}_i \rangle = 0$ we have

$$\langle \mathbf{x}_U, \mathbf{u}_i \rangle = \langle \mathbf{x} - \mathbf{x}_{U^\perp}, \mathbf{u}_i \rangle = \langle \mathbf{x}, \mathbf{u}_i \rangle - \langle \mathbf{x}_{U^\perp}, \mathbf{u}_i \rangle = \langle \mathbf{x}, \mathbf{u}_i \rangle,$$

hence $a_i = \langle \mathbf{x}, \mathbf{u}_i \rangle / \langle \mathbf{u}_i, \mathbf{u}_i \rangle$.

¹Here we use the convention that $V = V \oplus \{\mathbf{0}\}$.

Challenge Problem (Gram-Schmidt Procedure). Let V be a finite-dimensional inner product space over $\mathbb{R}$. The *Gram-Schmidt procedure* is used to convert any basis $\mathbf{v}_1, \dots, \mathbf{v}_n$ into an orthogonal basis $\mathbf{u}_1, \dots, \mathbf{u}_n$. We inductively define

$$\begin{aligned}\mathbf{u}_1 &:= \mathbf{v}_1, \\ \mathbf{u}_2 &:= \mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1, \\ \mathbf{u}_3 &:= \mathbf{v}_3 - \frac{\langle \mathbf{v}_3, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1 - \frac{\langle \mathbf{v}_3, \mathbf{u}_2 \rangle}{\langle \mathbf{u}_2, \mathbf{u}_2 \rangle} \mathbf{u}_2, \\ &\text{etc.}\end{aligned}$$

Afterwards, we may obtain an *orthonormal* basis by replacing each $\mathbf{u}_i$ with $\mathbf{u}_i/\|\mathbf{u}_i\|$.

- (a) Use induction to prove that the following statement holds for all $k \geq 1$:

$$\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\} \text{ and } \mathbf{u}_k \neq \mathbf{0}.$$

[Remark: We need $\mathbf{u}_k \neq \mathbf{0}$ so we can divide by $\langle \mathbf{u}_k, \mathbf{u}_k \rangle$ in the definition of $\mathbf{u}_{k+1}$.]

- (b) Use induction to prove that the following statement holds for all $k \geq 1$:

$$\langle \mathbf{u}_i, \mathbf{u}_k \rangle = 0 \quad \text{for all } i < k.$$

- (c) Define the subspace $U_k := \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$. Use Problem 4 to show that

$$\mathbf{u}_{k+1} = P_{U_k^\perp}(\mathbf{v}_{k+1}) \quad \text{for all } k \geq 1.$$

The idea: We create a new vector $\mathbf{u}_{k+1}$ that is perpendicular to the previous vectors $\mathbf{u}_1, \dots, \mathbf{u}_k$ by projecting the original $\mathbf{v}_{k+1}$ onto the orthogonal complement $U_k^\perp$. Parts (a) and (b) show that this idea works.

- (d) The Gram-Schmidt procedure can also be applied in infinite-dimensional vector spaces. Consider the function space $L^2[0, 1]$ with inner product

$$\langle f(x), g(x) \rangle = \int_0^1 f(x)g(x) dx.$$

If we apply Gram-Schmidt to the infinite sequence $1, x, x^2, \dots$, then we obtain an infinite sequence of polynomials $p_0(x), p_1(x), p_2(x), \dots$ called the *shifted Legendre polynomials*. Compute the first three Legendre polynomials and check that they are orthogonal with respect to the inner product.

- (e) Let $U = \text{Span}\{1, x, x^2\} \subseteq L^2[0, 1]$. Continuing from part (d), we know that

$$U = \text{Span}\{p_0(x), p_1(x), p_2(x)\}.$$

Use the formula from Problem 4 to compute the projection $f(x) := P_U(x^3)$ onto the subspace U . Draw the graphs of x^3 and $f(x)$ on the same axes.

- (a): To save space we define the notation

$$\begin{aligned}U_k &:= \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}, \\ V_k &:= \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}.\end{aligned}$$

We will use induction to show that the following statement holds for all $1 \leq k \leq n$:

$$U_k = V_k \quad \text{and } \mathbf{u}_i \neq \mathbf{0} \text{ for } 1 \leq i \leq k.$$

The statement holds for $k = 1$ because $\mathbf{u}_1 = \mathbf{v}_1$ and $\mathbf{v}_1 \neq \mathbf{0}$. (No basis contains the zero vector.) Now suppose that the statement holds for some $1 \leq k < n$. In this case we need to show that $U_{k+1} = V_{k+1}$ and $\mathbf{u}_{k+1} \neq \mathbf{0}$. Since $\langle \mathbf{u}_i, \mathbf{u}_i \rangle \neq 0$ for all $1 \leq i \leq k$ we may define

$$\mathbf{u}_{k+1} := \mathbf{v}_{k+1} - \sum_{i=1}^k \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \mathbf{u}_i.$$

In order to show that $V_{k+1} \subseteq U_{k+1}$ we will show that $\mathbf{v}_1, \dots, \mathbf{v}_{k+1} \in U_{k+1}$. We have $\mathbf{v}_{k+1} \in U_{k+1}$ because

$$\mathbf{v}_{k+1} = \mathbf{u}_{k+1} + \sum_{i=1}^k \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \mathbf{u}_i,$$

and we have $\mathbf{v}_1, \dots, \mathbf{v}_k \in U_{k+1}$ because $\mathbf{v}_1, \dots, \mathbf{v}_k \in V_k$ and $V_k = U_k \subseteq U_{k+1}$.

In order to show that $U_{k+1} \subseteq V_{k+1}$ we will show that $\mathbf{u}_1, \dots, \mathbf{u}_{k+1} \in V_{k+1}$. We have $\mathbf{u}_1, \dots, \mathbf{u}_k \in V_{k+1}$ because $\mathbf{u}_1, \dots, \mathbf{u}_k \in U_k$ and $U_k = V_k \subseteq V_{k+1}$. And we have $\mathbf{u}_{k+1} \in V_{k+1}$ because

$$\mathbf{w} := \sum_{i=1}^k \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \mathbf{u}_i \in U_k = V_k \subseteq V_{k+1},$$

hence $\mathbf{u}_{k+1} = \mathbf{v}_{k+1} - \mathbf{w} \in V_{k+1}$. It remains to show that $\mathbf{u}_{k+1} \neq \mathbf{0}$. But if $\mathbf{u}_{k+1} = \mathbf{0}$ then

$$\mathbf{v}_{k+1} = \sum_{i=1}^k \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \mathbf{u}_i \in U_k = V_k,$$

which contradicts the fact that the set $\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{v}_{k+1}$ is independent. (Any subset of an independent set is independent.)

(b): We have shown that $U_n = V_n = V$, so that $\mathbf{u}_1, \dots, \mathbf{u}_n$ is a spanning set. To complete the proof we need to show that

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0 \quad \text{for all } i \neq j.$$

We will do this by showing that the following statement holds for all $1 \leq k \leq n$:

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0 \quad \text{for all } 1 \leq i < j \leq k.$$

For the case $k = 1$ there is nothing to check. Suppose the statement holds for some $1 \leq k < n$, so that $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for all $1 \leq i < j \leq k$. Then for any $1 \leq i \leq k$ we also have

$$\begin{aligned} \langle \mathbf{u}_i, \mathbf{u}_{k+1} \rangle &= \left\langle \mathbf{u}_i, \mathbf{v}_{k+1} - \sum_{j=1}^k \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_j \rangle}{\langle \mathbf{u}_j, \mathbf{u}_j \rangle} \mathbf{u}_j \right\rangle \\ &= \langle \mathbf{u}_i, \mathbf{v}_{k+1} \rangle - \sum_{j=1}^k \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_j \rangle}{\langle \mathbf{u}_j, \mathbf{u}_j \rangle} \langle \mathbf{u}_i, \mathbf{u}_j \rangle \\ &= \langle \mathbf{u}_i, \mathbf{v}_{k+1} \rangle - \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \langle \mathbf{u}_i, \mathbf{u}_i \rangle \\ &= 0. \end{aligned}$$

Thus $\mathbf{u}_1, \dots, \mathbf{u}_n$ is an orthogonal basis for V .

(c): There is not much to show. Since $\mathbf{u}_1, \dots, \mathbf{u}_k$ is an orthogonal basis for the subspace $U_k \subseteq V$, Problem 4 tells us that

$$P_{U_k^\perp}(\mathbf{v}_{k+1}) = \mathbf{v}_{k+1} - \sum_{i=1}^k \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \mathbf{u}_i.$$

But the right hand side is exactly the definition of $\mathbf{u}_{k+1}$.

(d): Consider the subspace $U = \text{Span}\{1, x, x^2\} \subseteq L^2[0, 1]$. We can apply the Gram-Schmidt procedure to the basis $1, x, x^2$ to obtain an orthogonal basis $p_0(x), p_1(x), p_2(x)$ for U as follows. First we define

$$p_0(x) = 1.$$

Then we define

$$\begin{aligned} p_1(x) &= x - \frac{\langle x, p_0 \rangle}{\langle p_0, p_0 \rangle} p_0(x) \\ &= x - \frac{\int_0^1 x \, dx}{\int_0^1 1 \, dx} \cdot 1 \\ &= x - \frac{1}{2}. \end{aligned}$$

Next,

$$\begin{aligned} p_2(x) &= x^2 - \frac{\langle x^2, p_0 \rangle}{\langle p_0, p_0 \rangle} p_0(x) - \frac{\langle x^2, p_1 \rangle}{\langle p_1, p_1 \rangle} p_1(x) \\ &= x^2 - \frac{\int_0^1 x^2 \, dx}{\int_0^1 1 \, dx} \cdot 1 - \frac{\int_0^1 x^2(x - \frac{1}{2}) \, dx}{\int_0^1 (x - \frac{1}{2})^2 \, dx} \left(x - \frac{1}{2}\right) \\ &= x^2 - \frac{1}{3} - \frac{1/12}{1/12} \left(x - \frac{1}{2}\right) \\ &= x^2 - x + \frac{1}{6}. \end{aligned}$$

Thus the first three shifted Legendre polynomials are

$$p_0(x) = 1, \quad p_1(x) = x - \frac{1}{2}, \quad p_2(x) = x^2 - x + \frac{1}{6}.$$

We check that they are orthogonal:

$$\begin{aligned} \langle p_0, p_1 \rangle &= \int_0^1 \left(x - \frac{1}{2}\right) \, dx = 0, \\ \langle p_0, p_2 \rangle &= \int_0^1 \left(x^2 - x + \frac{1}{6}\right) \, dx = \frac{1}{3} - \frac{1}{2} + \frac{1}{6} = 0, \end{aligned}$$

and

$$\begin{aligned} \langle p_1, p_2 \rangle &= \int_0^1 \left(x - \frac{1}{2}\right) \left(x^2 - x + \frac{1}{6}\right) \, dx \\ &= \int_0^1 \left(x^3 - \frac{3}{2}x^2 + \frac{2}{3}x - \frac{1}{12}\right) \, dx \\ &= \frac{1}{4} - \frac{1}{2} + \frac{1}{3} - \frac{1}{12} \\ &= 0. \end{aligned}$$

(e): We can use the orthogonal basis $p_0(x), p_1(x), p_2(x)$ for the space U of quadratic polynomials to compute the orthogonal projection of any function in $L^2[0, 1]$ onto the subspace U . The projection $P_U(f)$ is the quadratic polynomial that is closest to f in the L^2 norm.

For the example $f(x) = x^3$, we have

$$P_U(x^3) = \frac{\langle x^3, p_0 \rangle}{\langle p_0, p_0 \rangle} p_0 + \frac{\langle x^3, p_1 \rangle}{\langle p_1, p_1 \rangle} p_1 + \frac{\langle x^3, p_2 \rangle}{\langle p_2, p_2 \rangle} p_2.$$

We compute

$$\frac{\langle x^3, p_0 \rangle}{\langle p_0, p_0 \rangle} = \frac{\int_0^1 x^3 dx}{\int_0^1 1 dx} = \frac{1}{4},$$

and

$$\frac{\langle x^3, p_1 \rangle}{\langle p_1, p_1 \rangle} = \frac{\int_0^1 x^3(x - \frac{1}{2}) dx}{\int_0^1 (x - \frac{1}{2})^2 dx} = \frac{3/40}{1/12} = \frac{9}{10}.$$

Finally,

$$\langle x^3, p_2 \rangle = \int_0^1 x^3 \left(x^2 - x + \frac{1}{6} \right) dx = \frac{1}{6} - \frac{1}{5} + \frac{1}{24} = \frac{1}{120},$$

and

$$\langle p_2, p_2 \rangle = \int_0^1 \left(x^2 - x + \frac{1}{6} \right)^2 dx = \frac{1}{180}.$$

Hence

$$\frac{\langle x^3, p_2 \rangle}{\langle p_2, p_2 \rangle} = \frac{1/120}{1/180} = \frac{3}{2}.$$

Therefore

$$\begin{aligned} f(x) := P_U(x^3) &= \frac{1}{4} + \frac{9}{10} \left(x - \frac{1}{2} \right) + \frac{3}{2} \left(x^2 - x + \frac{1}{6} \right) \\ &= \frac{3}{2} x^2 - \frac{3}{5} x + \frac{1}{20}. \end{aligned}$$

Thus

$$f(x) = \frac{3}{2} x^2 - \frac{3}{5} x + \frac{1}{20}.$$

Here are the graphs of x^3 and $f(x)$:

